

# Benjamin Johnston

(614) 843-3974 | [bejaunst@gmail.com](mailto:bejaunst@gmail.com) | [linkedin.com/in/bejaunst](https://linkedin.com/in/bejaunst) | [github.com/cetio](https://github.com/cetio) | [cetio.github.io](https://cetio.github.io)

## TECHNICAL SKILLS

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**Languages:** C, C++, C#, D, x86 Assembly, Python

**Systems & RE:** Reverse Engineering, ARM64/x86 Disassembly, Firmware Analysis, PE Format, CoreCLR Internals, FFI/Interop, Template Metaprogramming

**Security:** Vulnerability Research, Hook Detection, Windows Syscalls, Shellcode Generation, HMAC, CVE Analysis

**Performance:** SIMD/AVX-512, SSE, BMI2, AArch64 NEON, Custom Allocators, Cache-friendly Design

**Infrastructure:** Linux, Git, CMake, Docker, CI/CD

## EXPERIENCE

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### Software Engineer

Jul 2021 – Present

*Independent client engagements*

- Reverse-engineered the VEX Robotics V5 private firmware API from ARM64 disassembly, documenting 8,650 lines of C/C++ headers across 39 files with 481 firmware memory offsets; cited by the Rust vex-sdk crate.
- Designed and maintain 10+ open-source libraries in C, C#, C++, and D (700+ commits), spanning runtime introspection (Cassowary: 15 CoreCLR structures), Windows syscall research (UndeadInterop: 8 hook detection classifications), and a compile-time multi-ISA assembler (Multidasm: 651 AMD64 instructions).
- Contributed 120 commits to fern-pl/fnc (Fern Native Compiler), serving as the second-highest contributor to an upstream compiler project.

### Automation Engineer

Nov 2025 – Present

*Handshake*

- Built AutoHandshake!, a cross-platform (Linux, macOS, Windows) Selenium automation suite with unattended execution and UI-stall recovery, eliminating 95% of manual effort across 400+ runtime hours.

### AI Evaluation Engineer

Dec 2025 – Jun 2026

*Snorkel AI*

- Built Docker-based reproducible evaluation sandboxes with strict schema constraints, catching defects pre-release and enforcing training-set integrity before pipeline ingestion.

## SELECTED PROJECTS

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**VEXAPI** | *Firmware Reverse Engineering, C/C++* | [github.com/cetio/VEXAPI](https://github.com/cetio/VEXAPI)

- Reverse-engineered the VEX Robotics V5 private firmware API from ARM64 disassembly of VEXos using Ghidra, documenting 8,650 lines of C/C++ headers across 39 files with 481 firmware memory offsets.
- Produced 34 public API headers covering 20+ device types and 5 private headers with jump table and boot sequence signatures; cited by the Rust vex-sdk crate.

**UndeadInterop** | *Windows Syscall Research, C#* | [github.com/cetio/UndeadInterop](https://github.com/cetio/UndeadInterop)

- Resolved Windows syscall IDs from PE export tables and invoked them directly via runtime-generated x86\_64 shellcode stubs, bypassing ntdll usermode hooks.
- Implemented 8 hook detection classifications (inline jumps/calls, stack pivots, debug interrupts, forwarded exports) via Iced x86 decoder, with a complete NTSTATUS enum (339 values).

**Multidasm** | *Multi-ISA Assembler, D* | [github.com/cetio/multidasm](https://github.com/cetio/multidasm)

- Built a compile-time assembler targeting x86, x86\_64, CIL, and MIPS, with 651 AMD64 instruction template instantiations (79% coverage) and 226 CIL opcodes (100% coverage).
- Implemented SIMD instruction encoding (MMX through AVX2/FMA) with label resolution, patching, and block-based API with executable memory emission.

**Cassowary** | *.NET Runtime Introspection, C#* | [github.com/cetio/Cassowary](https://github.com/cetio/Cassowary)

- Authored a managed C# library for .NET runtime introspection, modeling 15 CLR internal structures (MethodTable, EEClass, MethodDesc, FieldDesc) and 15 type signature models.
- Implemented bi-directional C++ interop with unsafe pointer-based access and Reflection.Emit delegate construction; referenced by Novus, one of the largest .NET reflection libraries.

## EDUCATION

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**Columbus State Community College**

Expected Dec 2026

*AAS, Software Development*